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PHILOSOPHY

Two Boxes — The Argument That Never Settled

Two rules of rational choice that had agreed on everything suddenly disagreed about a million dollars, and fifty-seven years of professional argument has not decided which one is right.

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Somewhere on a table in front of you there are two boxes. The transparent one holds a thousand dollars; you can see the notes. The opaque one holds either a million dollars or nothing at all, and which of those it holds was decided yesterday, by a predictor who studied you and made a forecast. If the predictor forecast that you would take only the opaque box, it put the million in. If it forecast that you would take both boxes, it left the opaque box empty. The predictor has now gone. Nothing you do can change what is inside. You may take just the opaque box, or you may take both.

The problem was devised in 1960 by William Newcomb, a theoretical physicist at the Lawrence Livermore Laboratory who worked on magnetic confinement of plasma and never published a word about boxes. It travelled by word of mouth for most of a decade, through physicists and mathematicians, before Robert Nozick wrote it up for a volume of essays honouring Carl Hempel in 1969 [1]. Nozick opened with an observation that has aged better than any of the arguments that followed it: to almost everyone it is perfectly clear and obvious what should be done, and yet these people divide almost evenly, each half convinced the other is simply being silly. When Martin Gardner put the problem to the readers of *Scientific American* in July 1973, it drew one of the largest volumes of mail his column ever received [2].

Fifty-seven years later there is still no settled answer. That is unusual. Philosophy has plenty of open questions, but this one is not open in the way that consciousness or moral realism is open. It is a question about arithmetic and money, with no hidden vagueness in the terms, and two rules of rational choice that agree about every ordinary decision give it flatly opposite answers.

Two Rules That Had Never Disagreed

The first rule is dominance. If one option is at least as good as another in every possible state of the world, and strictly better in some, take it. Lay the four combinations out and dominance is not subtle.

What you do	Predictor forecast one box	Predictor forecast two boxes
Take the opaque box only	1,000,000	0
Take both boxes	1,001,000	1,000

Whatever the predictor did yesterday, taking both boxes pays exactly a thousand dollars more than taking one. Not on average — in each row of the table, with certainty. The money is already in the box or it is not, and reaching for the transparent box cannot reach backwards in time to remove it.

The second rule is expected utility, in the form Richard Jeffrey gave it in *The Logic of Decision* in 1965 [3]. Jeffrey's innovation was to build the whole theory on one algebra of propositions and to have an agent maximise *conditional* expected utility — the desirability of an act being the average of the outcomes weighted by their probability given that the act is performed:

$$V(A) = \sum_s P(s \mid A) u(A \wedge s)$$

Read the probabilities as news. Learning that you took one box is very good news about what is in the opaque box, because that is precisely the news the predictor's accuracy makes it. Learning that you took both is very bad news. Suppose the predictor is right nine times in ten. Then one-boxing has an expected value of 900,000 and two-boxing an expected value of 101,000, and the recommendation is not close.

Both rules are respectable. Neither had ever needed to yield to the other, because in an ordinary decision your act and the states you care about are related only through your act's effects, and then the two formulas coincide. Newcomb's problem is engineered so that they do not: your act is evidence about a state it cannot influence.

The repair that became orthodoxy was to make the causal structure explicit. Allan Gibbard and William Harper, in 1978, replaced the conditional probability with the probability of a counterfactual — the chance that *if I were to do A*, state *s* would obtain [4]:

$$U(A) = \sum_s P(A \Box \rightarrow s) u(A \wedge s)$$

That is causal decision theory, developed further by David Lewis into the version most philosophers learned [5]. Its verdict on the boxes is the same as dominance's, and for the same reason: your choice today does not counterfactually bear on a forecast made yesterday, so the probabilities are simply the unconditional ones and the extra thousand dollars is free. Jeffrey's original theory, now called evidential decision theory, says take one box. Both camps have had first-rate defenders — James Joyce's 1999 book is the standard modern statement of the causal view [15], Arif Ahmed's 2014 book the standard evidential reply [16] — and neither has dislodged the other.

The Predictor Barely Has to Try

BEFORE READING ON

The problem is usually posed with a predictor who is nearly infallible, which invites the suspicion that the whole thing rests on an impossible premise. So weaken it. How accurate does the predictor have to be before the expected value of taking one box exceeds the expected value of taking both?

Write B for the large prize, s for the small one, and p for the probability that the predictor gets this particular agent right. Taking one box returns B when the forecast was correct and nothing otherwise, so its expected value is pB . Taking both returns s when the forecast was correct and $B + s$ when it was wrong, which is $s + (1 - p)B$. Setting those equal gives a threshold that depends on the two prizes and nothing else:

$$p^* = \frac{B + s}{2B} = \frac{1,001,000}{2,000,000} = 0.5005$$

The predictor has to beat a fair coin by one-twentieth of a percentage point. That is the entire empirical demand the one-box argument makes on the world. A forecaster who is right 50.05 per cent of the time about how people handle this problem — a bar that psychologists clear routinely with far cruder instruments — is enough to put one-boxing ahead.

Predictor accuracy	One box	Both boxes	Difference
0.5000	500,000	501,000	-1,000
0.5005	500,500	500,500	0
0.7500	750,000	251,000	499,000
0.9900	990,000	11,000	979,000

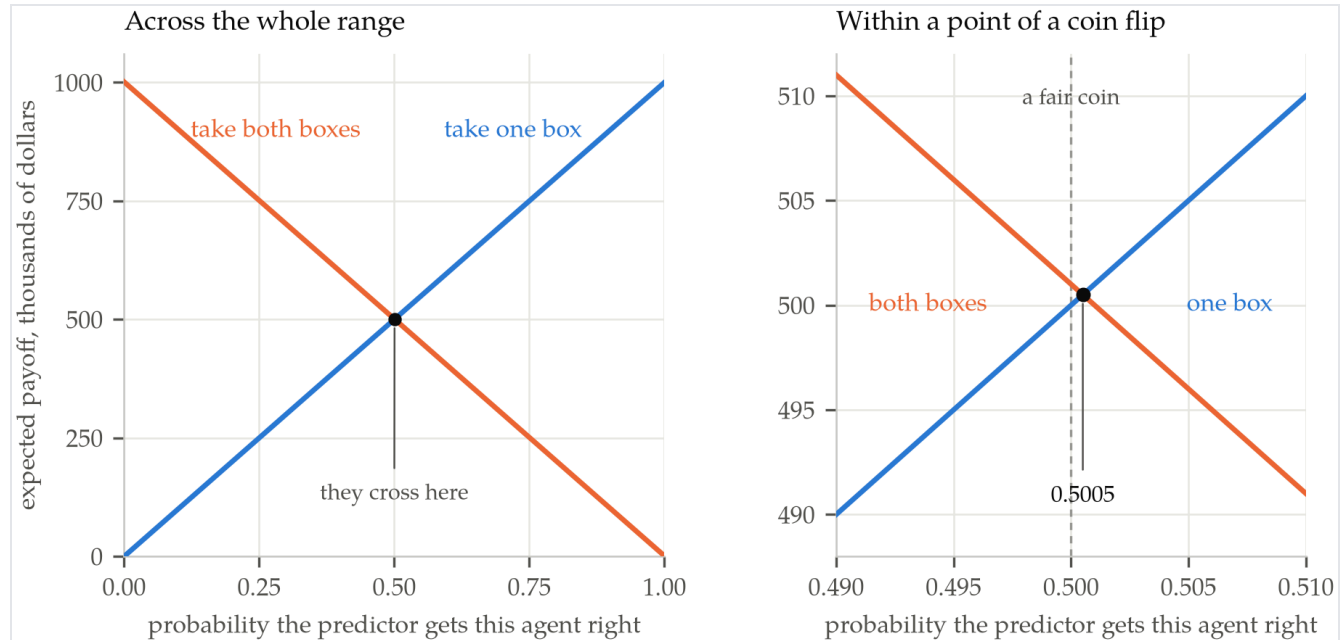


FIGURE Left: expected payoff from each choice as the predictor's accuracy runs from useless to perfect. Right: the same lines within a percentage point of a coin flip, the only window in which they are close enough to distinguish. The crossing sits 0.05 points above one half.

Away from that crossing the gap is enormous — at ninety-nine per cent accuracy, one-boxing is worth ninety times as much expected money. The two camps are not arguing about a marginal case. They are on opposite sides of a line that sits, for all practical purposes, at a coin flip, and one of them is defending a position that costs a thousandfold if the predictor is any good at all.

The Same Argument, in a Doctor's Office

The causal side's best move is not to defend the boxes directly but to show that the evidential rule, applied consistently, gives monstrous advice elsewhere. Consider a world in which smoking is powerfully correlated with lung cancer but does not cause it: a genetic lesion causes both the cancer and the craving. Should someone who enjoys smoking abstain? Evidential decision theory appears to say yes, because the news that you smoked is bad news about your genome — even though abstaining cannot alter a gene you already have.

This is not a purely invented case. It is close to the position Ronald Fisher argued in earnest. In a letter to *Nature* in August 1958 he reported that smoking habits are more alike in identical than in fraternal twins, including twins reared apart, and concluded that the propensity to smoke is partly constitutional, so that a genotype predisposing to both smoking and cancer could not be ruled out [7]. Fisher was wrong about the world, and the epidemiology went the other way, but he was not confused about the logic: correlation between a habit and a disease is compatible with either causal story, and the two stories license different advice.

Evidential theorists answered with what became known as the tickle defence. If the lesion works by producing a craving, then an agent who introspects on the craving already has the evidence; conditional on knowing how much she wants to smoke, the act itself carries no further news about the gene. Ellery Eells pushed this furthest, arguing that a sufficiently self-aware deliberator always has such a “tickle” available [8]. Whether that rescues the theory in general, or only in cases where the common cause happens to act through a felt desire, is still contested.

Then, thirty years on, the counterexamples started running the other way. Andy Egan's psychopath button asks you to imagine a button that will kill all psychopaths, which you would like, except that only a psychopath would press it, and if you are a psychopath the button kills you too [9]. Causal decision theory reasons that pressing cannot change whether you are a psychopath and endorses pressing — which is very nearly a proof that you will die. Worse for the theory's self-image is Death in Damascus, from Gibbard and Harper's original paper [4]: Death will meet you tomorrow in whichever city you go to, and has already predicted your move, so whichever city causal reasoning recommends, it immediately recommends the other. The theory is not merely wrong here; it is unstable, giving advice it retracts the moment you accept it. Frank Arntzenius made that instability the organising problem of decision theory and argued that a rational agent should never be able to foresee regretting her own choice [10].

Case	Causal	Evidential	Functional
Newcomb's boxes	Both boxes	One box	One box
Smoking lesion	Smoke	Abstain	Smoke
Psychopath button	Press	Refrain	Refrain
Death in Damascus	Unstable	Unstable	Stable choice

The third column is Functional Decision Theory, proposed by Benjamin Levinstein and Nate Soares in 2020 [11]. Instead of asking which physical act would produce the best outcome, it asks which *output of one's decision procedure* would — treating every implementation of the same function as moving together, for logical rather than causal reasons. It gets the whole column above right, which is more than either older theory manages, at the price of a dependence relation that is neither causation nor correlation and that its critics regard as unexplained.

Fifty Years of Not Converging

The disagreement is measurable, because philosophers have been polled. In the 2009 PhilPapers survey, 292 of 931 respondents took both boxes and 198 took one, with the rest declining to answer — 31.4 against 21.3 per cent of the full sample, or roughly sixty-forty among those who committed [13]. The 2020 repeat, covering about two thousand philosophers, put it at 39.0 per cent for two boxes and 31.2 for one [12]. Eleven years and a large literature moved the margin by less than two points.

A DIRECTION WORTH CHECKING

Suppose you were told that one subgroup of philosophers holds a far more lopsided view of the boxes than the profession at large. Which subgroup, and which way?

Specialists in decision theory, and toward two boxes: 61 per cent against 26 [12]. This is the single most awkward datum in the field. The people who have thought hardest about the problem lean hardest toward the option that, on the problem's own stipulations, predictably collects a thousand dollars while the other camp collects a million. Almost every other area of philosophy leans the other way.

That embarrassment has a name. “If you're so smart, why ain'cha rich?” is the one-boxer's taunt, and Lewis answered it in three pages in 1981: one-boxers do end up richer, but they are not rewarded for rationality, they are rewarded for irrationality, because the predictor pays out for a disposition rather than for good reasoning [14]. Whether that is a defence or a confession has been argued ever since. Nozick himself never fully settled; by 1993 he was proposing that rational decision maximises a weighted sum of causally and evidentially expected utility, with a third term for the symbolic value of an act, which concedes that neither pure rule is adequate [23].

A newer data point arrives from an unexpected direction. In 2024 Caspar Oesterheld and colleagues hand-built a dataset of several hundred natural-language questions about Newcomb-like problems and scored large language models on them. Models' tendency to one-box correlated with their general capability and with their decision-theoretic competence: the stronger the model, the more evidential its instincts [17]. Whatever that shows about the boxes, it is a curious inversion of the specialists' verdict.

Two Games in One Costume

There is a third response, less popular than either camp and in some ways more convincing: that the question is not fully asked. David Wolpert and Gregory Benford argued in 2013 that the two recommendations are not rival answers to one question but correct answers to two different questions, because the problem statement never fixes the joint probability distribution over your choice and the forecast [18]. Write the situation as a Bayes net and there are two ways to complete it. Their "Fearful" player takes the forecast to be probabilistically informative about the choice and computes accordingly; their "Realist" treats the choice as independent of the forecast, so that conditioning on it changes nothing. Both are coherent. The verbal statement of Newcomb's problem is consistent with either, and each camp has been quietly supplying the missing structure in its own favour and then finding the other camp obtuse.

Judea Pearl reaches a similar place from causal inference, with a sharper edge. The distinction the dispute turns on is the one between observing that an action occurred and intervening to make it occur — between $P(y | x)$ and $P(y | \text{do}(x))$ — and Pearl's verdict is that confusing actions with acts is what produced Newcomb's paradox in the first place [19]. That is a clean diagnosis and also where the argument restarts, because the disputed question is exactly whether a deliberating agent should model her own choice as an intervention from outside the system or as one more event inside it, correlated with everything that produced her.

Where the Boxes Are Real

None of this would matter much if the structure were confined to thought experiments. It is not. The signature of a Newcomb problem is an act that is diagnostic of an outcome it does not cause, and social life is full of them.

George Quattrone and Amos Tversky demonstrated in 1984 that people reason evidentially about themselves whether or not they should. In one experiment, subjects held a forearm in circulating cold water, were told that either increased or decreased tolerance after exercise indicated a healthy heart and long life, and then shifted their tolerance in the diagnostic direction — while mostly denying they had done any such thing. In a second, subjects told that an election would be decided by which party's supporters turned out were markedly more willing to vote, and more inclined to believe their own

turnout would bring out others like them, than subjects told the outcome hung on unaligned voters. Quattrone and Tversky named it the voter's illusion [20]. One vote among millions has no causal purchase on an election; it is excellent evidence about how people who resemble you will behave.

Macroeconomics contains the same structure in a form nobody calls philosophical. Finn Kydland and Edward Prescott showed in 1977 that a policymaker who re-optimises every period, facing agents who forecast policy rationally, ends up worse off than one bound to a rule — the optimal plan is not time-consistent, and discretion delivers inflation without output [21]. The private sector is the predictor; the policy choice is the boxes; and the recommendation to *commit*, to give up the free thousand dollars of period-by-period flexibility, is one-boxing with a central bank's balance sheet behind it. That the same logic won a Nobel Prize in economics while remaining unresolved in philosophy is worth noticing.

Douglas Hofstadter ran the experiment on his own colleagues. In 1983 he posed a twenty-person prisoner's dilemma to a group of mostly scientists and mathematicians, telling each of them plainly that all the others were about equally bright and reasoning the same way, and argued that agents who know they are alike should recognise their choices as one choice — what he called superrationality. Fourteen of the twenty defected [22].

Now put two artificial agents in a negotiation, each with the other's weights, or the same model instantiated twice. Their choices are not causally linked and are not statistically independent either; they are the same computation running in two places. That is Newcomb's problem with the predictor replaced by a copy, and it is the practical reason the Levinstein–Soares proposal and the model evaluations sit in the same literature [11], [17]. A design question about machine cooperation now depends on which side of a 1969 argument you take.

Newcomb's problem is best read as a diagnostic rather than a puzzle. It separates two things ordinary life keeps welded together: what an action brings about, and what it reveals about the kind of agent taking it. Most of the time both point at the same act, which is why one rule sufficed for so long. When they come apart, the question stops being which rule is correct and becomes which of two projects rationality is for — steering outcomes, or being the sort of agent that good outcomes accrue to. Bar-Hillel and Margalit saw it in 1972, observing that dominance loses its grip precisely where the states of the world are not independent of the chooser [6]. Fifty-seven years has produced no winner because there is no fact of the matter to find, only a choice about what a theory of choice is meant to do. The boxes are still on the table, and the disagreement about which to take is the most durable result in the field.

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